

(iii) $U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S}) = 0$ for some prior b' for agent i . Then $U_i^{(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})}(\mathcal{S}) \leq U_i^{(\mathbf{b}^*, \mathbf{b}^{**})}(\mathcal{S}) = 0$ implies that for an agent with prior b' playing $\mathcal{I}$ is a (weakly) better response than sharing when other agents play according to $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$. By monotonicity of $U_i(\mathcal{S})$ in prior b_i , this implies that $b_i^{*,BR} \leq \hat{b}_i^{*,BR}$. If $\tilde{b} < b'$, then $b_i^{*,BR} = \hat{b}_i^{*,BR} = \tilde{b}$; otherwise, if $\tilde{b} \geq b'$, $b_i^{*,BR} = b_i^{**,BR} = b' \leq \hat{b}_i^{*,BR}$.

This establishes that $(\hat{\mathbf{b}}_i^{*,BR}, \hat{\mathbf{b}}_i^{**,BR}) \succeq (\mathbf{b}_i^{*,BR}, \mathbf{b}_i^{**,BR})$, so the order $\succeq$ is preserved by ψ . ■

Lemma A.3. *An increase in polarization of beliefs can always be constructed via the following process: take every belief b_i and either (i) add some $\epsilon_i > 0$ to b_i if $b_i > 1/2$, or (ii) subtract some $\epsilon_i > 0$ to b_i if $b_i < 1/2$.*

Proof of Lemma A.3. Let H_2 be more polarized than H_1 . For part (i), note that $H_1(b_i^1) = \alpha > 1/2$, so by single-crossing at $H_1^{-1}(1/2) = H_2^{-1}(1/2)$, we know that $H_2^{-1}(\alpha) - H_1^{-1}(\alpha) > 0$. Thus, for some $b_i^2 > b_i^1$, we have $H_2^{-1}(\alpha) = b_i^2$, or in other words, $H_2(b_i^2) = \alpha$. Setting $\epsilon_i = b_i^2 - b_i^1 > 0$ in this fashion for all $b_i > 1/2$ accomplishes claim (i). For part (ii), note that $H_1(b_i^1) = \alpha < 1/2$, so by single-crossing at $H_1^{-1}(1/2) = H_2^{-1}(1/2)$, we know that $H_2^{-1}(\alpha) - H_1^{-1}(\alpha) < 0$. Thus, for some $b_i^2 < b_i^1$, we have $H_2^{-1}(\alpha) = b_i^2$, or in other words, $H_2(b_i^2) = \alpha$. Setting $\epsilon_i = b_i^1 - b_i^2 > 0$ in this fashion for all $b_i < 1/2$ accomplishes claim (ii). ■

Lemma A.4. *If $\kappa \leq \bar{\kappa} \equiv (c\tilde{u} - u(\tilde{u} - \tilde{c})) / (\tilde{u}N)$, then for any agent i :*

- (i) *If $b_i^* > 0$ and $b_i^{**} < 1$, then $b_i^{**} > b_i^*$;*
- (ii) *For all $\bar{b} < 1$, there exists $\tilde{r} > 0$ such that agent i plays $\mathcal{D}$ in any equilibrium for an article with $r < \tilde{r}$ and on any sharing network $\mathbf{P}$, provided that $b_i < \bar{b}$.*

Proof of Lemma A.4. For part (i), by way of contradiction suppose that $b_i^* = b_i^{**}$. Then for an agent with prior b_i^* (and corresponding ex-post belief π_i^* that the article is truthful), it must be the case that:

$$\tilde{u}(1 - \pi_i) - \tilde{c} = u\pi_i - c(1 - \pi_i) + \mathbb{E}[\kappa S_i - dD_i] \geq 0.$$

Re-arranging we get that $\pi_i = \frac{\tilde{u} - \tilde{c} + c - \mathbb{E}[\kappa S_i - dD_i]}{\tilde{u} + u + c}$. Plugging back into the payoff for action $\mathcal{D}$, we see that:

$$U_i(\mathcal{D}) = \tilde{u} \left(\frac{u + \tilde{c} + \mathbb{E}[\kappa S_i - dD_i]}{\tilde{u} + u + c} \right) - \tilde{c} \leq \tilde{u} \left(\frac{u + \tilde{c} + \kappa N}{\tilde{u} + u + c} \right) - \tilde{c} < \tilde{u} \left(\frac{u + \tilde{c} + \bar{\kappa} N}{\tilde{u} + u + c} \right) - \tilde{c} \leq 0.$$

By assumption, $U_i(\mathcal{S}) = U_i(\mathcal{D}) < 0$, but since $U_i(\mathcal{I}) = 0$, ignoring is the best response at prior b_i^* , which is a contradiction.

For part (ii), notice by equation (2), for a fixed $b < 1$, as $r \rightarrow 0$, $\pi_i \rightarrow 0$, and therefore:

$$U_i(\mathcal{S}) = u\pi_i - c(1 - \pi_i) + \mathbb{E}[\kappa S_i - dD_i] < u\pi_i - c(1 - \pi_i) + \bar{\kappa}N \leq u\pi_i - c(1 - \pi_i) + \frac{c}{N}N \stackrel{r \rightarrow 0}{\rightarrow} -c + c = 0.$$

where the last inequality follows from the observation that:

$$\bar{\kappa} \equiv \frac{c\tilde{c} - u(\tilde{u} - \tilde{c})}{\tilde{u}N} < \frac{c\tilde{c}}{\tilde{u}N} < \frac{c}{N},$$